

Deep Learning theory - Lecture 8

Summary of "classical" statistical learning theory:

$$\text{let } R_* = \inf_{f: X \rightarrow Y} R(f)$$

$$\begin{aligned} \Delta(A, \mathcal{D}_n) &= R(\hat{\sigma}) - \inf_{f: X \rightarrow Y} R(f) = \underbrace{R(\hat{\sigma}) - R_*}_{\text{approx} \rightarrow 0 \text{ as } p \rightarrow \infty} \\ &\quad + \underbrace{R(\hat{\sigma}) - \hat{R}(\hat{\sigma})}_{\text{gen}} + \underbrace{\hat{R}(\hat{\sigma}) - \hat{R}(\hat{\sigma}_*)}_{\substack{\text{est} \\ \rightarrow 0 \text{ as } t \rightarrow \infty \\ \text{convex}}} + \underbrace{R(\hat{\sigma}_*) - R(\hat{\sigma}_*)}_{\substack{\text{gen} \\ \rightarrow 0 \text{ if } n \rightarrow \infty \\ \text{faster than } p}} \end{aligned}$$

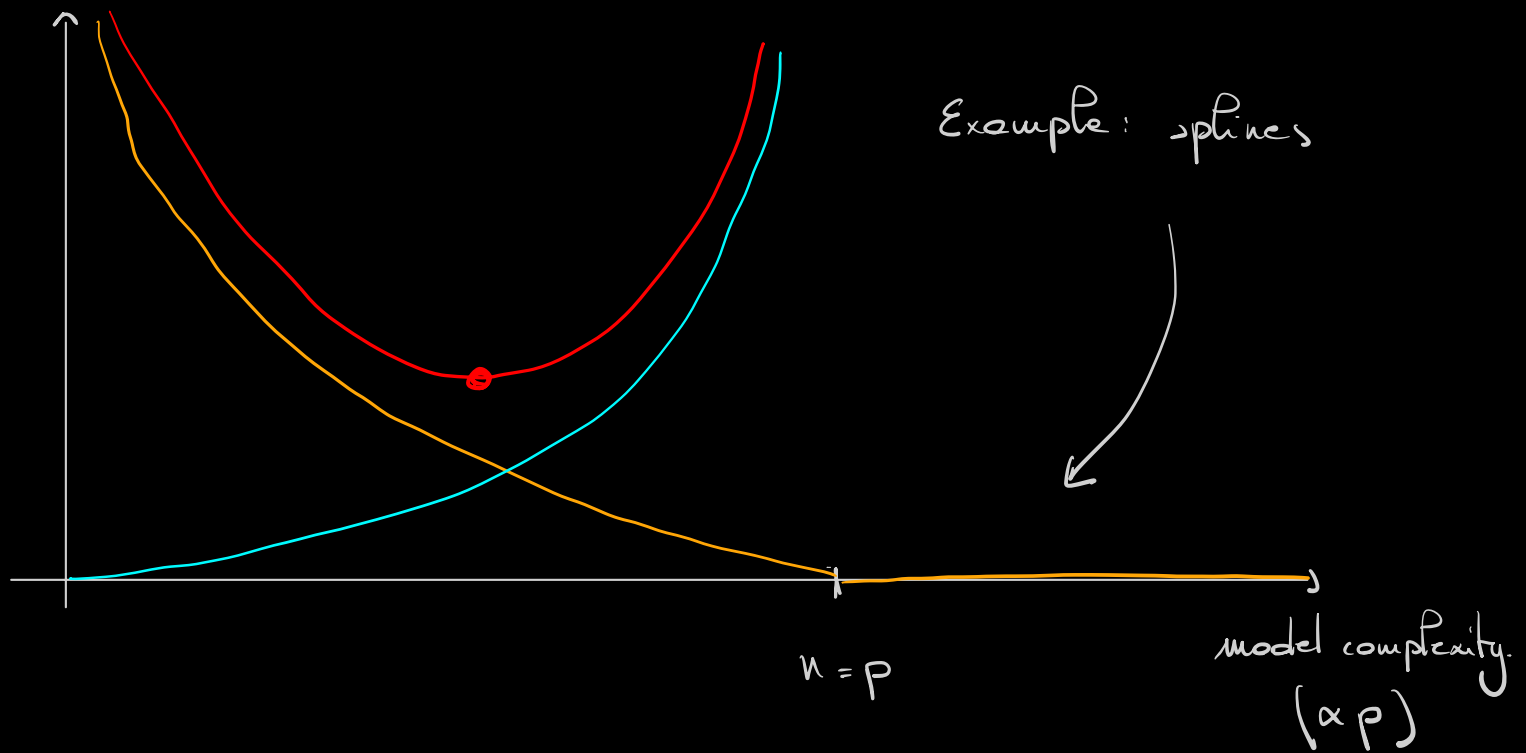
Example: (Bias/variance tradeoff for MSE).

$$\text{Consider } \ell(f_\theta(x), y) = \frac{1}{2} \|f_\theta(x) - y\|^2 \implies R = \frac{1}{2n} \sum_{j=1}^n \|f_\theta(x_j) - y_j\|^2$$

$$y_j = f_*(x_j) + \varepsilon_j \quad \varepsilon_j \sim \mathcal{N}(0, \sigma^2)$$

$$\begin{aligned} \mathbb{E}_{\mathcal{D}_n} (R(\hat{\sigma}) - \inf_f R(f)) &= \mathbb{E} \left(\frac{1}{2n} \sum_i \|f_{\hat{\sigma}}(x_i) - y_i\|^2 \right) - \mathbb{E} \left(\frac{1}{2n} \sum_i \|f - f_i\|^2 \right) \\ &= \frac{1}{2n} \sum_i \left[\mathbb{E} (f_{\hat{\sigma}}(x_i))^2 - 2 \mathbb{E} (f_{\hat{\sigma}}(f_* + \varepsilon)) + \mathbb{E} (f_* + \varepsilon)^2 \right] \cdot \sigma^2 \\ &= \frac{1}{2} \mathbb{E} [f_{\hat{\sigma}}^2] - [\mathbb{E} (f_{\hat{\sigma}})]^2 + \underbrace{(\mathbb{E} (f_{\hat{\sigma}} - f_*))^2}_{\text{Bias}(f_{\hat{\sigma}})^2} + \underbrace{\mathbb{E} (f_*^2)}_{\text{Var}(f_{\hat{\sigma}})} + \sigma^2 \\ &\quad \underbrace{\mathbb{E}_{\mathcal{D}_n} (f_{\hat{\sigma}} - \mathbb{E}_{\mathcal{D}_n} (f_{\hat{\sigma}}))^2}_{\text{Var}(f_{\hat{\sigma}})} = \text{Var}(f_{\hat{\sigma}}) + \text{Bias}(f_{\hat{\sigma}})^2 \end{aligned}$$

provided that n is large enough we expect the following behavior



How does one find the red point?

(Tikhonov) Regularization:

One can penalize models with large values of σ (complex) by adding a term to the loss function ℓ that penalizes that extent:

$$R(\sigma) = \frac{1}{n} \sum \ell(f_\sigma(x_i), y_i) + \lambda \underbrace{\|f_\sigma\|_H}_{\text{(Tikhonov) regularization term}}$$

Example (linear regression)

Consider minimizing $R(\beta) = \frac{1}{2n} \|X \cdot \beta - y\|^2 + \lambda \|\beta\|_2^2$

The minima are given by: $\frac{1}{n} X^T (X \beta_* - y) + \lambda \mathbb{1}_d \beta_* = 0$

$$\Rightarrow \beta_* = \left(\frac{1}{n} X^T X + \lambda \mathbb{1}_d \right)^{-1} (X^T y)$$

Fact: empirically, this is not the regime where NN work:

- NN work in a regime where the above assumptions do not hold:

- nonconvexity: The risk $R(\theta)$ is nonconvex for NN:

consider for instance

$$f_{\theta}(x) = a_1 \sigma(w_1 x) + a_2 \sigma(w_2 x) \quad \theta = (a_1, w_1, a_2, w_2)$$

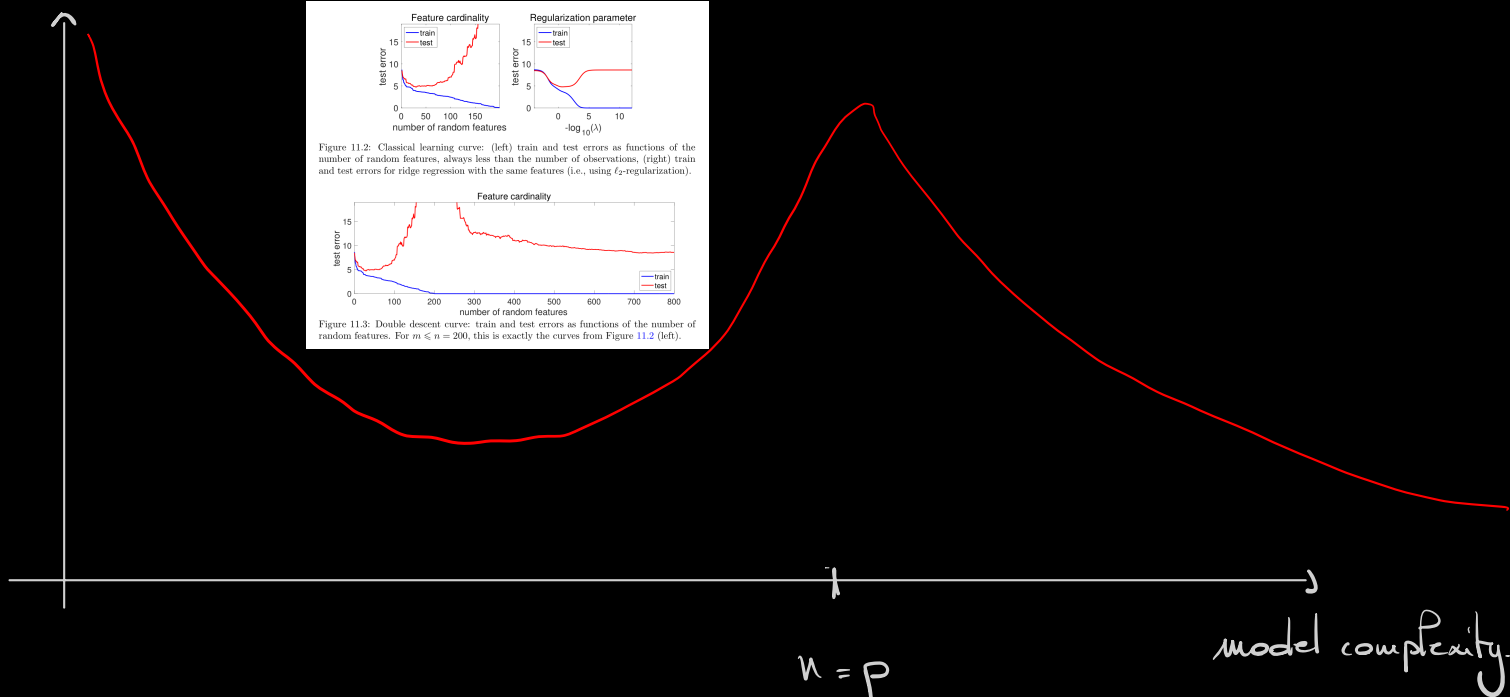
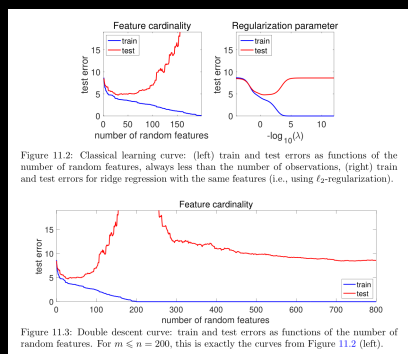
$$\text{learning } y = f_{\pi}(x) = 1 \cdot \sigma(1 \cdot x) + \sqrt{2} \cdot \sigma(\pi \cdot x)$$

both $\theta_1 = (1, 1, \sqrt{2}, \pi)$ and $\theta_2 = (\sqrt{2}, \pi, 1, 1)$ yield $R(\theta) = 0$

but $\theta_3 = \frac{\theta_1 + \theta_2}{2} = \left(\frac{1+\sqrt{2}}{2}, \frac{1+\pi}{2}, \frac{1+\sqrt{2}}{2}, \frac{1+\pi}{2}\right)$ gives $R(\theta_3) > 0$

$\Rightarrow$ nonconvex in θ !

Further, NN often work in the p > n regime.



How can we explain the performance of NN?

Infinite-width limits for neural networks.

Consider a single-layer NN given by:

$$f_{\theta}(x) = \frac{1}{\sqrt{m}} \sum_{j=1}^m a_j \sigma(w_j \cdot x + b_j)$$

where the parameters are initialised iid according to arbitrary distributions f_a, f_w, f_b .

Notes: - throughout, we will assume that $\mathbb{E}(a) = 0$

- concretely, people initialise the network using

$$f_{\theta}(x) = \sum \tilde{a}_j \sigma(w_j x + b_j)$$

$w_j \stackrel{\text{iid}}{\sim} f_w, b_j \stackrel{\text{iid}}{\sim} f_b$ and $\tilde{a}_j \stackrel{\text{iid}}{\sim} \tilde{f}_a$ with $\text{Var}(\tilde{a}_j) = \frac{1}{m} \text{Var}(a_j)$

- The output of the network at initialisation converges to a gaussian process in the limit $m \rightarrow \infty$:

Since $\mathbb{E}_{\theta} f_{\theta}(x) = 0 \quad \forall x$, consider the random vector

$$Y = \begin{pmatrix} f_{\theta}(x_1) \\ \vdots \\ f_{\theta}(x_n) \end{pmatrix} = \frac{1}{\sqrt{m}} \sum_{j=1}^m \begin{pmatrix} f_{(w_j, a_j, b_j)}(x_1) \\ \vdots \\ f_{(w_j, a_j, b_j)}(x_n) \end{pmatrix} \xrightarrow[m \rightarrow \infty]{\text{CLT}} \mathcal{N}(0, \Sigma)$$

$$\Sigma_{x, x'} = \mathbb{E}_{f_a, f_w, f_b} (f_{(w, a, b)}(x) f_{(w, a, b)}(x')) = 0.0$$

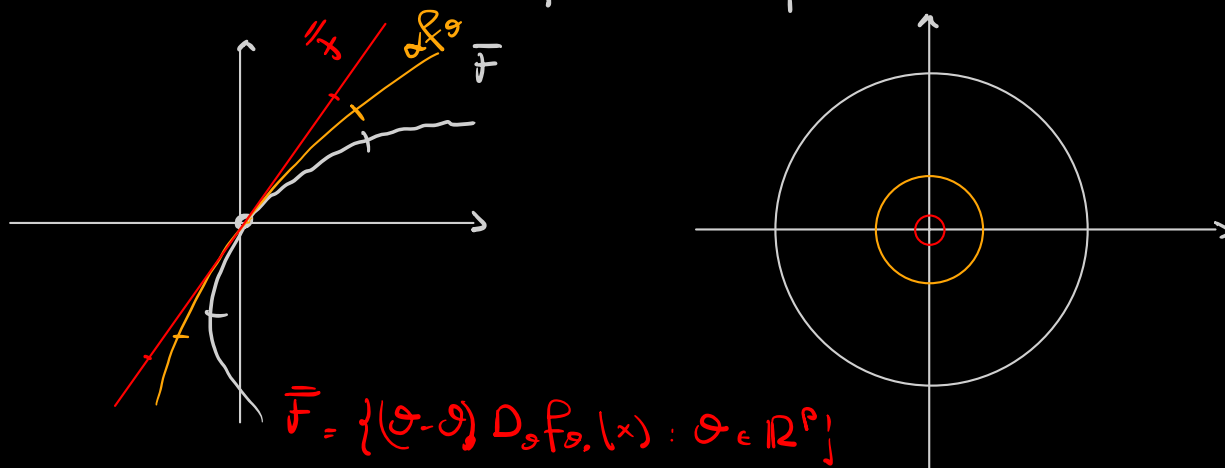
Question: How do NN (NTK) behave as $n \rightarrow \infty$ during training?

Heuristic discussion: consider rescaling a model f_θ as

$$f_\theta \rightarrow \alpha f_\theta \quad \text{for } \alpha \gg 1.$$

To make sense of initialization, assume $f_{\theta_0} \equiv 0$.

Then, the model in function space looks like



$$\bar{F} = \{(\theta - \theta_0) D_\theta f_{\theta_0}(x) : \theta \in \mathbb{R}^p\}$$

$$\Rightarrow \bar{F}_\alpha \xrightarrow[\alpha \rightarrow \infty]{} \bar{F}$$

It turns out that neural networks in (NTK) are in this regime

None of this is rigorous, but the intuition can help us construct a proof

Consider the MSE $\hat{R}(\theta) = \frac{1}{2n} \sum (y_i - f_\theta(x_i))^2 = \frac{1}{2n} \|\vec{y} - \vec{F}(\theta)\|_2^2$

where $\vec{F}_n(\theta) = (f_\theta(x_1) \dots f_\theta(x_n))$.

The GF evolution of the model is given by

$$\frac{d}{dt} \theta_t = \frac{1}{n} D_{\theta} \vec{F}_n(\theta) (\vec{y} - \vec{F}_n(\theta)). \quad D_{\theta} \vec{F}_n \in \mathbb{R}^{n \times p}$$

we want to compare this with the evolution of the linearized model in $\vec{F} := \{ \vec{F} = \vec{f}_{\vartheta_0}(x) + D_{\vartheta} \vec{f}_{\vartheta_0}(x) (\vartheta - \vartheta_0) : \vartheta \in \mathbb{R}^p \}$.

$$\frac{d}{dt} \vec{\vartheta}_t = \frac{1}{n} D_{\vartheta} \vec{F}_n(\vartheta_0) \cdot (\vec{y} - \vec{F}_n(\vartheta_0) - D_{\vartheta} \vec{F}_n(\vartheta_0) (\vec{\vartheta}_t - \vartheta_0))$$

which is the GF for the ish

$$\hat{R}_{\text{lin}}(\vec{\vartheta}) = \frac{1}{2n} \|\vec{y} - \vec{F}_n(\vartheta_0) - D_{\vartheta} \vec{F}_n(\vartheta_0) (\vec{\vartheta} - \vartheta_0)\|_2^2$$

Rem: if $D_{\vartheta} \vec{F}_n(\vartheta_0)$ has full row rank, the set

$$\text{ERM}_0 = \{ \vec{\vartheta} : \hat{R}_{\text{lin}}(\vec{\vartheta}) = 0 \} \text{ is dim} = p - n \text{ affine sp.}$$

$$\text{and GF } \vec{\vartheta}_t \xrightarrow{t \rightarrow \infty} \vec{\vartheta}_{\infty} = \arg \min \{ \|\vec{\vartheta} - \vartheta_0\|^2 : \vec{F}_n(x) = y \}.$$

To quantify deviations from linearity we consider

$$\text{Lip}(D_{\vartheta} \vec{F}) = \sup_{\vartheta_1 \neq \vartheta_2} \frac{\|D_{\vartheta_1} \vec{F}_n(\vartheta_1) - D_{\vartheta_2} \vec{F}_n(\vartheta_2)\|}{\|\vartheta_1 - \vartheta_2\|}$$

where $\|A\| = \sup_{v \in \mathbb{R}^p, \|v\|_2=1} \|Av\|_2$ and the population version

$$\text{Lip}(D_{\vartheta} f(\vartheta)) = \frac{\|D_{\vartheta_1} f(\vartheta_1) - D_{\vartheta_2} f(\vartheta_2)\|}{\|\vartheta_1 - \vartheta_2\|_2}$$

where $\|D_{\vartheta} f(\vartheta)\| = \sup_{v \in \mathbb{R}^p} \frac{\|Df(\vartheta)v\|_{L^2(P)}}{\|v\|_2}$ for $Df: \mathbb{R}_+^p \rightarrow L^2(P)$

Thm: Let $\sigma_{\max} = \sigma_{\max}(Df_u(\vartheta_0))$, $\sigma_{\min} = \sigma_{\min}(Df_u(\vartheta_0))$

Assume $\text{Lip } D_{\vartheta} \bar{f}_u \|y - \bar{f}_u(\vartheta_0)\| \leq \frac{1}{4} \sigma_{\min}^2$

Then for all $t \geq 0$:

$$1) \quad \hat{R}(\vartheta_t) \leq \hat{R}(\vartheta_0) e^{-\lambda_0 t} \quad \lambda_0 = \sigma_{\min}^2 / 2u$$

$$2) \quad \|f(\vartheta_t) - \bar{f}(\vartheta_t)\|_{L^2(\rho)} \leq \left(\frac{\text{Lip}(Df)}{\sigma_{\min}^2} + \frac{\|Df(\vartheta_0)\| \text{Lip } Df_u}{\sigma_{\min}^5 / \sigma_{\max}^2} \right) \cdot \|y - \bar{f}_u(\vartheta_0)\|^2$$

Lemma: • $\text{Lip}(Df) = O(\sqrt{d})$ $\text{Lip}(Df_u) = O(\sqrt{d}(\sqrt{n} + \sqrt{d}))$

$$\bullet \sigma_{\min} = O(\sqrt{d}) \quad \sigma_{\max} = O(\sqrt{n} + \sqrt{d})$$

$$\bullet \|y - f_u(\vartheta_0)\| = O(\sqrt{n}) \quad \|Df(\vartheta_0)\| = O(1)$$

Thm: Let $(x_i, y_i)_{i=1}^n$: $x_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \mathbb{I}_d)$, y_i $\mathbb{R}^2$ -sub-gaussian.

Let $\sigma = \sum_{e \in \mathcal{C}_0} \mu_e(\sigma) h_e$ for h_e the Hermite polynomials, and $\mu_e(\sigma) \neq 0 \quad \forall \quad e \in \mathcal{C}_0$. Further let $\ln d \leq \ln \log u$, $u \leq d^{e_0}$.

Then w.p. at least $1 - \exp(-u/c)$, for all $t \geq 0$ we have

$$1) \quad \hat{R}(\vartheta_t) \leq \hat{R}(\vartheta_0) e^{-\lambda_* t} \quad \lambda_* = C_1 \frac{d}{u}$$

$$2) \quad \|f_{\vartheta_t} - \bar{f}_{\vartheta_t}\|_{L^2(\rho)} \leq C_1 \left(\sqrt{\frac{u^2}{nd}} + \sqrt{\frac{u^5}{nd^4}} \right) = O\left(\frac{1}{\sqrt{n}}\right)$$